Student Marks Predictor Using PyTorch

AIM

To demonstrate a basic machine-learning model that learns from sample study-hour and marks data and predicts marks for a given number of study hours.

REQUIREMENTS

Python 3, PyTorch, and NumPy. Install packages in Mac Terminal using: python3 -m pip install torch numpy

PROCEDURE

1. Import PyTorch.

2. Create sample data relating study hours to marks.

3. Create a simple linear model.

4. Train the model for 1000 iterations using a loss calculation and an optimizer.

5. Give the model 4 study hours and print its predicted marks.

PROGRAM CODE

```
import torch

model = torch.nn.Linear(1, 1)

x = torch.tensor([[1.0], [2.0], [3.0]])
y = torch.tensor([[20.0], [40.0], [60.0]])

optimizer = torch.optim.SGD(model.parameters(), lr=0.01)

for i in range(1000):
    prediction = model(x)
    loss = ((prediction - y) ** 2).mean()

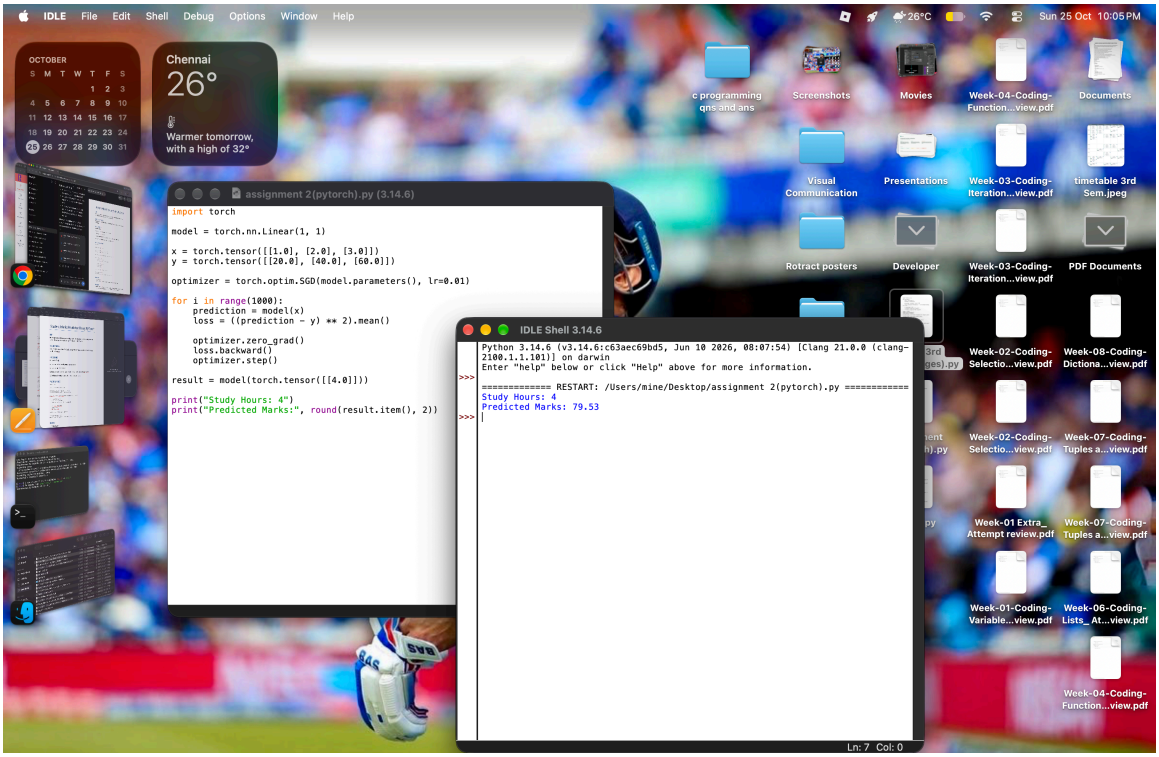
    optimizer.zero_grad()
    loss.backward()
    optimizer.step()

result = model(torch.tensor([[4.0]]))

print("Study Hours: 4")
print("Predicted Marks:", round(result.item(), 2))
```

OUTPUT

Study Hours: 4
Predicted Marks: approximately 80 (the exact value may vary slightly).



RESULT

A simple PyTorch model was trained on sample data and used to predict marks for four study hours. This is a classroom demonstration, not a reliable real-world marks predictor.